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Performance, Power Efficiency and Scalability of Asymmetric Cluster Chip Multiprocessors

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Abstract—**This paper evaluates asymmetric cluster chip multiprocessor (ACCMP) architectures as a mechanism to achieve the highest performance for a given power budget. ACCMPs execute serial phases of multithreaded programs on large high-performance cores whereas parallel phases are executed on many small and simple cores. Theoretical analysis reveals a performance upper bound for symmetric multiprocessors, which is surpassed by asymmetric configurations at certain power ranges. Our emulations show that asymmetric multiprocessors can reduce power consumption by more than two thirds with similar performance compared to symmetric multiprocessors.**

Index Terms—**ACCMP, CMP.**

I. INTRODUCTION

ACHIEVING high performance within a given power envelope is a major concern for microprocessor architects. In the past, the constant decrease in feature sizes has enabled to increase uniprocessor performance by packing more transistors in the same die area. Nowadays, due to the complexity of current state of the art microprocessors, a large increase in power and area results in only small performance improvements. According to Pollack's empirical rule [7], performance of uniprocessors is proportional to the square root of their area. These trends favor the use of Chip Multi Processors (CMP) [5].

Software applications contain serial phases as well as parallel phases. The lowest execution time for the serial phases is achieved by the highest performance uniprocessor available, since the serial phases can be executed on one processor only. On the other hand, parallel phases can be executed on numerous processors in parallel. Therefore, the lowest execution time for the parallel phases is achieved by executing them on many simple processors that consume less energy per instruction (EPI) [4]. We claim that a choice of symmetric cores is suboptimal due to the contradicting requirements of the serial and parallel phases within the same application.

We propose placing clusters of different cores on a single die. All of the cores within the Asymmetric Cluster Chip Multi Processor (ACCMP) share the same instruction set architecture and memory address space in order to let threads migrate from core to core. The larger high-performance cores will execute single-threaded programs and the serial phases of multithreaded programs. The smaller cores, though each providing lower performance than their larger counterparts, will execute the parallel phases at better performance versus power ranges. In the general case, ACCMPs will include clusters of different cores, since numerous multithreaded applications with varying parallelism execute in parallel. An example of an ACCMP with three clusters is illustrated in Fig. 1.